

Structure:

1. Introduction — 400 words Open with the non-stationarity thesis: the 2026–27 El Niño is not arriving into the climate system of 1997 or 2015. Warming baselines are shifting the loss distributions that conventional models are calibrated against. Quantify the Niño 3.4 anomaly, place it in historical context with ensemble spread, and state the analytical claim: expected losses will be systematically underestimated by models conditioned on pre-2020 historical analogs.

El Niño is arriving into a different climate system.

The 2026–27 event is not unfolding against the baselines of 1997–98 or 2015–16. Decades of anthropogenic warming have raised mean tropical sea surface temperatures, shifted loss distributions, and amplified the extremes that conventional catastrophe models — calibrated predominantly on pre-2020 historical analogs — are built to price.¹

As of mid-June 2026, the weekly Niño 3.4 index stands at approximately $+0.7^{\circ}\text{C}$ ², with recent pentad values nearing $+0.9^{\circ}\text{C}$, confirming the onset of El Niño conditions. At comparable points in the development of 1997–98 and 2015–16, onset anomalies of similar magnitude preceded seasonal peaks well above 2.0°C by boreal winter. Current ensemble guidance from the NMME and IRI plume assigns a 97–98% likelihood of El Niño persistence through the Northern Hemisphere winter (NDJ),³ indicating near-certainty of continued event development. More significantly, CPC's official ENSO Strength Probabilities — derived from the 3-month seasonal Relative Oceanic Niño Index averaged across the Niño-3.4 region — assign a 63% probability that the NDJ seasonal average will reach or exceed $+2.0^{\circ}\text{C}$, the threshold for a very strong event.⁴ This would place 2026–27 alongside 1982–83, 1997–98, and 2015–16 as among the strongest events in the observational record. This trajectory is not reassuring.

Conventional frameworks, anchored in stationary assumptions from cooler historical periods, will systematically underestimate expected losses from this event.⁵ They insufficiently capture non-stationary teleconnection responses — heavier precipitation extremes in flood-prone regions and deeper droughts elsewhere — superimposed on a baseline

¹ Moore, F. (2024). *Learning, Catastrophic Risk and Ambiguity in the Climate Change Era*. National Bureau of Economic Research (NBER); Abrams, J. F. (2026). "Recalibrating Climate Risk: Aligning Damage Functions with Scientific Understanding." *Green Futures Solutions*. <https://greenfuturesolutions.com/wp-content/uploads/2026/01/Recalibrating-Climate-Risk-Report-Final.pdf>

² NOAA Climate Prediction Center, "ENSO Diagnostic Discussion," National Oceanic and Atmospheric Administration, https://www.cpc.ncep.noaa.gov/products/analysis_monitoring/enso_advisory/ensodisc.shtml (accessed June 13, 2026).

³ International Research Institute for Climate and Society, "ENSO: Recent Evolution, Current Status and Predictions," Columbia Climate School, <https://iri.columbia.edu/our-expertise/climate/forecasts/enso/current/> (accessed June 13, 2026).

⁴ NOAA Climate Prediction Center, "Official NOAA CPC ENSO Strength Probabilities," National Oceanic and Atmospheric Administration, https://www.cpc.ncep.noaa.gov/products/analysis_monitoring/enso/roni/strengths/ (accessed June 13, 2026).

⁵ European Central Bank (2025). *Climate-related risk assessment and the challenge of non-stationary tails*. [replace with specific ECB publication URL before submission]

approximately 1.0°C warmer than the 1997 reference period. They also miss correlated multi-basin loss structure, as concurrent drought in Southeast Asia and flooding in South America produce supply-chain cascades that break the independence assumptions embedded in standard reinsurance pricing.⁶ The result, across correlated multi-peril portfolios, is underpriced exceedance probability curves and systematic shortfalls in capital reserving — a miscalibration this article develops across physical, modeling, and financial dimensions.

2. Physical Science Layer — 700 words Bjerknes feedback mechanics briefly. Then focus analytically on two primary teleconnection regions for 2026–27: SE Asia drought signal and southern US flood signal. Develop each with probabilistic outlook, historical analog comparison (1997–98, 2015–16), and compound event framing (drought plus heat, flood plus antecedent soil saturation). Brief reference to East Africa and NE Brazil. One visualization: multi-panel global teleconnection anomaly map with confidence shading.

Bjerknes Feedback Mechanics

At the heart of El Niño lies the Bjerknes feedback: a positive ocean-atmosphere coupling loop. Warmer sea surface temperatures (SSTs) in the central-eastern equatorial Pacific reduce the east-west SST gradient, weakening the Walker circulation and easterly trade winds. These weakened winds reduce upwelling of cold deep water and allow warm water to shift eastward, further amplifying the initial SST anomaly. In a warmer baseline climate, elevated background SSTs mean the Bjerknes feedback is activated against a higher thermal energy state — the same wind stress anomaly produces a larger SST response, and the same SST response produces more extreme precipitation and drought teleconnections than historical analogs reflect. This feedback interacts with elevated background SSTs, potentially intensifying extremes even if ENSO amplitude itself shows limited change.

Primary teleconnection case 1 — Southeast Asia drought signal

Structure this analytically rather than descriptively. Three components: the mechanism (El Niño suppresses the western Pacific warm pool convection, weakening the Asian summer monsoon and reducing precipitation across the Maritime Continent and mainland Southeast Asia), the probabilistic outlook for 2026–27 (IRI tercile precipitation forecasts showing below-normal probability exceeding 50% across most of Southeast Asia and northern Australia for JJA and SON 2026), and the compound event framing (the drought signal is being superimposed on antecedent soil moisture deficits from multi-year warming trends, meaning the same precipitation deficit produces larger agricultural and hydrological impacts than historical distributions reflect). Close with the 1997–98 and 2015–16 analog: both events produced severe drought across the region, but current baseline soil moisture and temperature conditions amplify the compound stress beyond what those analogs captured.

Specific figures to include: historical El Niño precipitation anomalies for Southeast Asia of -1 to -3mm/day across the region during strong events, rice yield impacts of 3–8% in major producing countries (Thailand, Vietnam, Indonesia) during past strong El Niño years, and current IRI probabilistic outlook confidence levels.

⁶ Gaupp, F., Hall, J., Hochrainer-Stigler, S., and Dadson, S. "Changing Risks of Concurrent Food-Production Shocks in the Global Food System." *Nature Climate Change* 10 (2020): 54–57. <https://doi.org/10.1038/s41558-019-0629-0>.

Primary teleconnection case 2 — Southern US flood signal

Same three-component structure. Mechanism: El Niño strengthens the subtropical jet stream across the southern United States, increasing winter precipitation across the Gulf Coast, lower Mississippi Valley, and Southwest. Probabilistic outlook: IRI tercile precipitation forecasts showing above-normal probability exceeding 50% across the southern tier for DJF 2026–27. Compound event framing: increased precipitation is being superimposed on urban catchments where impervious surface coverage has expanded materially since 1997–98, producing higher runoff ratios, faster concentration times, and greater surface water flood extents than historical loss records reflect.

Historical analog: the 1997–98 event produced significant flood losses across the southern US — but infrastructure exposure and urban extent have grown substantially since then. FEMA National Flood Insurance Program loss data from 1997–98 and 2015–16 events provides the calibration baseline.

Secondary references — East Africa and NE Brazil

One paragraph combining both. East Africa typically receives above-normal rainfall during El Niño — the 1997–98 event produced devastating floods across Kenya, Somalia, and Ethiopia. NE Brazil experiences drought as the ITCZ shifts northward, suppressing convection over the semi-arid Nordeste region. Both effects are documented in IRI probabilistic outlooks for 2026–27. Keep this brief — one sentence each on mechanism, one on outlook, one on why these are secondary rather than primary analytical cases for this article's risk modeling focus.

Visualization guidance

Multi-panel global teleconnection anomaly map. Four panels: JJA 2026 precipitation anomaly probability (above/below normal tercile), SON 2026, DJF 2026–27, MAM 2027. Confidence shading based on IRI ensemble agreement. Source data: IRI probabilistic seasonal outlook maps, downloadable as GeoTIFF or accessible via IRI Data Library. Overlay historical analog composites from 1997–98 and 2015–16 as comparison panels. The visual argument is that 2026–27 ensemble guidance is tracking the upper end of historical strong event analogs across most teleconnection-active regions.

Section 3 — CAT Modeling Layer (800 words)

This is the section that most directly signals CAT modeling competency. Structure it in four parts.

Part 1 — The non-stationarity problem stated precisely

Open with the methodological argument rather than a general description of CAT modeling. The specific claim: ENSO-conditioned CAT models adjust near-term hazard rates by applying loss modifiers derived from historical event sets during comparable ENSO phases. The implicit stationarity assumption — that a given Niño 3.4 anomaly produces a comparable loss distribution regardless of the background climate state — is the failure

mode this section diagnoses. This is not a criticism of CAT modeling methodology per se. It is a calibration problem: the reference period from which loss modifiers are derived no longer represents the climate system the models are operating in.

Part 2 — EP curve construction and comparison

This is the concrete analytical demonstration the section requires. The procedure:

Obtain historical drought loss data for Southeast Asia from EM-DAT (Centre for Research on the Epidemiology of Disasters). Filter for flood and drought events in Thailand, Vietnam, Indonesia, Philippines, and Myanmar from 1980 to 2024. Normalize losses to 2024 USD using GDP deflators.

Obtain Niño 3.4 index data from NOAA CPC for the same period.

Run a regression of annual normalized losses on contemporaneous and lagged Niño 3.4 values. The coefficient on the Niño 3.4 term gives you the loss modifier per degree of anomaly — the same quantity that ENSO-conditioned CAT models are applying, but derived transparently from public data.

Construct two EP curves. The baseline EP curve uses the unconditional loss distribution from the full historical record. The El Niño-conditioned EP curve shifts the loss distribution upward by the regression-implied modifier for a very strong event ($\text{Niño } 3.4 \geq 2.0^\circ\text{C}$), with uncertainty bands derived from the regression confidence interval.

The climate-adjusted EP curve applies an additional upward shift reflecting the non-stationarity argument: losses in the warmest recent years (2015–2024) are systematically higher than the regression coefficient derived from the full record would predict, because the baseline warming amplifies the teleconnection response. Quantify this by comparing the regression residuals in the post-2015 period against the full-period residuals — the systematic positive residuals in recent years are your empirical evidence of non-stationarity.

The result should show three curves: baseline, ENSO-conditioned, and climate-adjusted. The gap between the ENSO-conditioned and climate-adjusted curves is your central analytical finding — the systematic underestimation embedded in conventional modeling.

Report the return period at which losses exceed a benchmark threshold (e.g., USD 5 billion aggregate SE Asian agricultural losses) under each curve. A loss level that is a 1-in-20-year event under the baseline may be a 1-in-10 or 1-in-8 year event under the climate-adjusted curve. This return period shift is the number your CAT modeling audience will find most legible.

Part 3 — Secondary peril: correlated multi-basin loss structure

The independence assumption in standard reinsurance pricing treats regional perils as having low or zero correlation across basins. El Niño systematically violates this assumption by simultaneously driving drought in Southeast Asia and flooding in South America from the same physical driver. Quantify the correlation: using the same EM-DAT dataset, calculate the correlation coefficient between Southeast Asian drought loss years and South American

flood loss years during strong El Niño phases versus neutral phases. The difference in correlation structure between ENSO phases is your empirical evidence that the independence assumption is inappropriate during strong El Niño events.

The reinsurance pricing implication: aggregate expected losses across a globally diversified agricultural or flood reinsurance book are higher during strong El Niño events than the sum of independently priced regional expected losses would imply. The magnitude of this diversification reduction is quantifiable from the correlation coefficient shift.

Part 4 — Forward-looking calibration requirements

Close the section with the methodological recommendation. ENSO-conditioned CAT models should incorporate three adjustments for 2026–27: recalibration of loss modifiers using post-2015 observational data rather than full historical records, explicit treatment of multi-basin loss correlation during El Niño phases, and scenario-weighted uncertainty bounds derived from NMME ensemble spread rather than point estimates of Niño 3.4 peak anomaly. The uncertainty range matters as much as the central estimate for capital reserving purposes — a wide ensemble spread implies fat tails in the loss distribution that point-estimate conditioning will miss.

Visualization

Three EP curves on a single chart: baseline, ENSO-conditioned, climate-adjusted. X-axis: return period (years, log scale). Y-axis: aggregate SE Asian agricultural loss (USD billions, 2024 normalized). Confidence bands on the climate-adjusted curve derived from regression uncertainty. Annotate the return period shift at a benchmark loss level. This is the article's most important single visualization — it makes the non-stationarity argument visible rather than merely argued.

Section 4 — Financial Implications Layer (900 words)

Three sub-sections in strict analytical hierarchy.

Sub-section 1 — Commodity price transmission

The regression structure: dependent variable is monthly log price change for soy, rice, and maize (FAO or World Bank commodity price data). Independent variable is contemporaneous and lagged Niño 3.4 index (1-month through 9-month lags). Run separate regressions for each commodity. Report the lag structure at which the coefficient is most statistically significant — this gives you the commodity-specific transmission timing that is the actionable insight for investment positioning.

Expected findings based on literature: maize and soy show significant response at 3–6 month lags, consistent with harvest cycle timing in major producing regions. Rice shows a longer and more variable lag, reflecting the role of strategic stockpiling and government intervention in rice markets, particularly in Southeast Asia. Wheat is less directly affected by ENSO but shows indirect effects through substitution dynamics when rice and maize prices spike.

Historical calibration: the 1997–98 event produced approximately 20–30% increases in global rice prices with a 6–9 month lag. The 2015–16 event produced more muted commodity price responses despite comparable Niño 3.4 anomalies, partly because global grain stocks were at historically high levels entering that event — a condition that does not currently apply. Current global grain stock-to-use ratios should be reported and compared against the 2015–16 entry point to establish that the commodity price transmission risk is higher this cycle.

Sub-section 2 — Equity sector analysis

The correlation analysis: using monthly sector index returns and Niño 3.4 index values, calculate rolling 12-month correlations between Niño 3.4 and returns for: agricultural ETFs (MOO — VanEck Agribusiness, CROP — IQ Global Agribusiness Small Cap), consumer staples (XLP), industrial/logistics (IYT), and reinsurance (specific names: Munich Re, Swiss Re, Hannover Re as proxies given limited pure-play reinsurance ETFs).

The lagged structure matters here too — equity markets partially anticipate commodity price moves, so correlations at 1–3 month lags between Niño 3.4 and sector returns may be more significant than contemporaneous correlations. Test both.

Identify the asymmetric risk positions: which sectors face the largest downside exposure and which face upside from a very strong El Niño? Agricultural input companies (fertilizer, irrigation equipment) in drought-affected regions face demand destruction. Agricultural commodity traders with diversified sourcing face margin expansion opportunities. Reinsurers with concentrated SE Asian agricultural book exposure face adverse loss development. Infrastructure companies in flood-resilient construction face increased demand in southern US markets.

The investment implication is not a prediction — it is a structured way of thinking about sector rotation and hedging under a specific climate scenario. Frame it explicitly as scenario analysis rather than as a forecast.

Sub-section 3 — Macro sovereign risk

Focus on three country clusters where ENSO-driven physical risks have the most direct sovereign credit implications.

Southeast Asian agricultural exporters: Thailand, Vietnam, and Indonesia are major rice and palm oil exporters. A severe El Niño-driven drought reduces export revenues, widens current account deficits, and can trigger food price inflation domestically — generating political economy pressures that complicate monetary policy. World Bank physical risk transmission estimates suggest GDP drag of 0.5–2.0% in severely affected agricultural export economies during strong El Niño years.

East African food importers: countries with high food import dependence and limited fiscal buffers face the most acute sovereign stress. Kenya, Ethiopia, and Somalia face simultaneous flood risk (above-normal rainfall) and the fiscal cost of humanitarian response. Sovereign spread widening during past El Niño events in this cluster provides the historical calibration.

Latin American commodity producers: Peru and Ecuador face direct flood and infrastructure damage. Brazil faces a more complex picture — the Amazon basin receives above-normal rainfall during El Niño while the Nordeste faces drought, creating regional divergence within a single sovereign. Soy production in the Cerrado region is less directly affected than Nordeste subsistence agriculture.

Visualization

Sub-section 1: commodity price response chart — line chart showing indexed price trajectories for rice, maize, and soy during 1997–98 and 2015–16 events, with 2026 baseline indexed to 100 and projected trajectory based on regression coefficients. Include confidence bands.

Sub-section 2: sector correlation heatmap — rows are sectors, columns are Niño 3.4 lag periods (0–9 months), cells are correlation coefficients color-coded from negative (blue) to positive (red). This makes the lag structure visible and identifies which sectors lead versus lag the physical signal.

Sub-section 3: sovereign exposure map — world map with countries colored by composite ENSO vulnerability score combining agricultural GDP share, food import dependence, fiscal buffer (reserves as percentage of GDP), and historical El Niño loss-to-GDP ratio.

Section 5 — Policy and Structural Implications (500 words)

The section's analytical spine is the crop insurance miscalibration argument. Develop it in three steps.

First, establish the mechanism: crop insurance actuarial tables in ENSO-exposed markets are typically calibrated on 20–30 year historical loss records. These records span a period of substantially lower mean temperatures than the current baseline. The loss frequencies and severities embedded in current premium structures do not reflect the non-stationary loss environment. This is not an argument about climate change in the abstract — it is a specific actuarial calibration problem with a measurable direction of error.

Second, develop two specific cases. SE Asian rice insurance: Thailand and Vietnam have expanding government-supported crop insurance programs. Premium structures are calibrated on historical drought loss frequencies that understate current risk under the warming baseline argument developed in Section 3. The EP curve shift quantified in that section directly translates into a premium adequacy gap — if the 1-in-20 event has become a 1-in-10 event, premiums calculated on 1-in-20 loss expectations are approximately half the actuarially required level. US Midwest drought coverage: USDA's Federal Crop Insurance program uses county-level historical yield data for premium calculation. The same non-stationarity argument applies — yield loss frequencies are increasing faster than the historical record captures, creating a systematic underpricing that transfers risk to the federal government rather than the market.

Third, connect to disclosure frameworks. TCFD requires financial institutions to assess physical risk under climate scenarios. NGFS scenario frameworks provide the standardized climate pathways against which portfolio exposure should be stress-tested. The

non-stationarity argument in this article — that ENSO-conditioned loss distributions are miscalibrated against warming baselines — is precisely the kind of physical risk miscalibration that TCFD disclosures are designed to surface. Regulators and institutional investors using TCFD-aligned disclosure to assess reinsurer and agricultural lender exposure should be asking whether the loss models underpinning those disclosures are subject to the calibration error this article identifies.

Close with the early warning investment argument: the physical ENSO signal is available 3–6 months before agricultural impacts materialise and 6–12 months before commodity price transmission. Investment in early warning systems — NOAA operational forecasting, WFP FEWS NET food security monitoring, national meteorological service capacity in exposed countries — has among the highest benefit-cost ratios of any climate adaptation investment precisely because the lead time is available. The 2026–27 event is a case study in whether that lead time is being used.

Section 6 — Conclusion (300 words)

Three paragraphs, each with a specific function.

Paragraph 1 — restate the non-stationarity thesis as the article's core contribution. The 2026–27 El Niño is not merely a strong climate event. It is a diagnostic test for whether risk modeling infrastructure has incorporated the structural shift in loss distributions that anthropogenic warming has produced. The evidence across physical science, CAT modeling, financial markets, and policy frameworks suggests it has not — systematically and in a directionally consistent way.

Paragraph 2 — the single most important indicator to monitor. The Niño 3.4 trajectory through boreal winter 2026–27 relative to NMME ensemble spread is the primary physical monitoring variable. But the more operationally important indicator for risk managers is the SE Asian agricultural growing season precipitation anomaly relative to ENSO composite baselines — this is the first point at which the physical signal translates into observable agricultural impact, and it provides a 2–3 quarter lead on the commodity price and equity market transmission.

Paragraph 3 — directional claim and closing. The largest underpriced risk in the current environment is SE Asian agricultural supply chain contagion into global food inflation — specifically the compounding of ENSO-driven crop shortfalls with existing supply chain fragility and low global grain stock-to-use ratios. For decision-makers in CAT modeling, the required action is EP curve recalibration using post-2015 observational data. For investment analysts, the required action is running lagged Niño 3.4 correlations against sector exposures now — the commodity price signal will arrive in 3–6 months. For policymakers, the intervention window is already open. Close with: "The ocean switched gears in June 2026. The question is whether our risk frameworks have."

Priority order for writing

Complete Section 2 first — the physical science layer is the foundation everything else rests on and it is currently the least developed. Then Section 3's EP curve construction, which requires the EM-DAT data acquisition and regression work. Sections 4, 5, and 6 can be written in sequence once 2 and 3 are complete. Come back with Section 2 when it is drafted and we will assess before moving to the CAT modeling layer.